AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Smart-FIX,Intelligent Machine Fault Diagnosis and Repair Recommendation

Team ID: *GEN26-A0H-I070*

Team Name: *INFINITE LOOPS*

Domain: *1 – Intelligent Systems and Digital Futures*

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Problem Statement

In automated industrial plants, a single machine fault can trigger several alarms at once. Technicians must manually compare sensor readings, equipment manuals, and past maintenance records to determine the underlying cause. This delays troubleshooting, prolongs production downtime, and increases maintenance and operating costs.

Problem to solve

How can we combine industrial machine data and maintenance knowledge to help technicians identify likely root causes faster and choose appropriate next diagnostic steps?

Problem Understanding

Who is affected?

Machine operators, maintenance technicians, troubleshooting engineers, automation engineers, shift supervisors, and production managers.

What happens today?

Technicians inspect alarms, review sensor readings, search documentation, compare earlier incidents, and then test a suspected cause.

Why does it happen?

Multiple alarms may represent one underlying fault and its symptoms. Sensor data, manuals, and maintenance records are often consulted separately.

Why it matters?

Diagnosis delays can extend downtime, interrupt production schedules, increase costs, and require unnecessary physical inspections.

Core distinction: detecting an abnormal condition is not the same as explaining its root cause.

Existing Gaps & Challenges

1. **Alarm overload:** several warnings may appear together, making the initiating fault difficult to separate from downstream symptoms.
2. **Fragmented information:** live sensor data, technical manuals, fault codes, and historical work orders may reside in different systems.
3. **Manual diagnosis:** technicians must connect the evidence themselves, and results can depend on individual experience and documentation quality.
4. **Limited actionable context:** a monitoring alert may identify an abnormal condition without presenting relevant evidence, past incidents, or safe next checks in one place.

Opportunity: connect machine-condition monitoring with context-aware root-cause investigation and technician decision support.

Our Proposed Solution

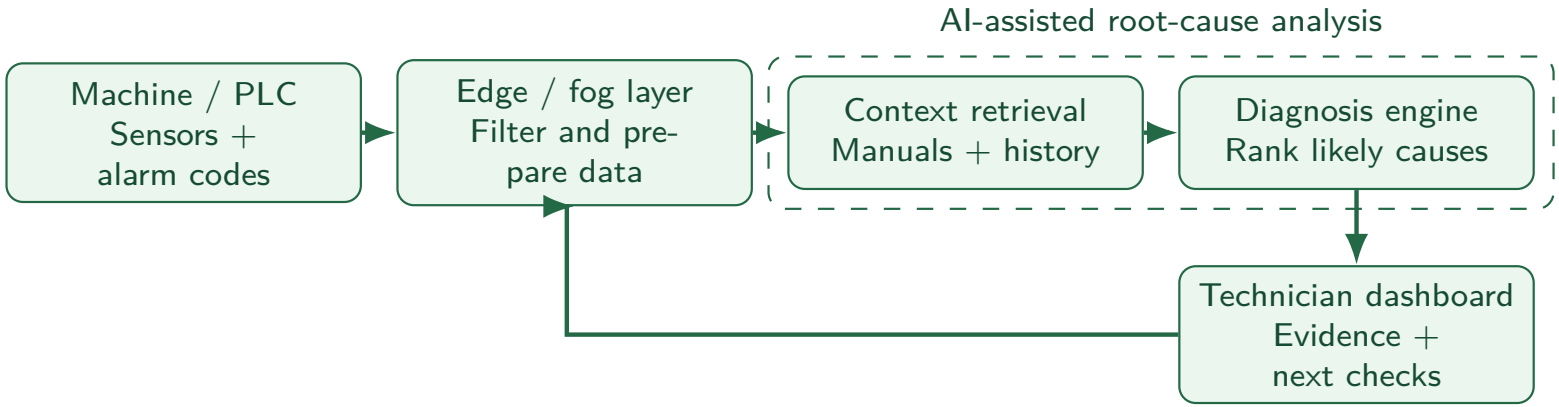
Proposed system

An AI-assisted Industrial IoT troubleshooting platform that brings together machine alarms, available sensor readings, equipment documentation, and historical maintenance records.

1. **Monitor:** collect available machine signals and alarm messages.
2. **Detect:** flag readings or operating conditions that appear abnormal.
3. **Contextualise:** retrieve relevant manual sections and similar past maintenance incidents.
4. **Diagnose:** rank plausible root causes using the combined evidence and explain why each is suspected.
5. **Recommend:** show suggested checks and supporting references for a qualified technician to verify.

The system supports technician decisions; it does not autonomously authorise hazardous repairs or replace machine safety interlocks.

Technical Approach



Proposed software stack

Python for data processing; a lightweight API/backend; a dashboard such as Streamlit; and SQLite or PostgreSQL for structured records.

Data and AI components

MQTT or an available plant interface where supported; rule-based/anomaly detection; document retrieval; and an LLM-assisted explanation layer grounded in retrieved sources.

Architecture is a proposed design for the team prototype. Final hardware, protocols, models, and deployment choices depend on available equipment and data. Research context: Khanna et al., "Preventive breakdown and fault detection of machine using industrial IOT in maintenance and automation" (2023).

Innovation & Uniqueness

Capability	Typical separated workflow	Our proposed approach
Monitoring	Alarms and readings viewed in monitoring tools	Use available live signals as diagnostic evidence
Context	Manuals and past work orders searched separately	Retrieve relevant documents and similar incidents together
Diagnosis	Technician manually connects symptoms	Rank plausible causes and show the evidence behind them
Action	Next checks depend on manual investigation	Suggest traceable checks for technician review

Key differentiator

From an alarm to ranked causes, supporting evidence, and suggested checks.

Differentiator: integrate monitoring data, maintenance knowledge, and explainable diagnostic assistance.

Feasibility & Scalability

Feasible prototype path

- Begin with sample or historical alarm and sensor data.
- Add a small, curated set of equipment manuals and maintenance records.
- Build a dashboard showing alarms, likely causes, evidence, and suggested checks.
- Validate outputs with a maintenance/automation subject-matter reviewer.

Scaling path

- Add machines through standardised data connectors.
- Separate plant-level edge processing from central reporting.
- Index manuals and work orders by machine, component, and fault code.
- Track feedback to improve retrieval and diagnosis quality.

Risks and mitigations: incomplete/noisy data (data-quality checks); incorrect AI suggestions (evidence citations and human approval); network/security constraints (read-only integration, access control, and IT/OT review); limited labelled failures (start with rules and expert-validated examples).

Cost for Product (Indicative Prototype Budget)

Costs depend on whether factory sensors, a gateway, and computing resources are already available. The following is a planning framework, not a vendor quotation.

Cost component	Planning estimate	Assumption
Existing laptop / development PC	INR 0 incremental	Reuse available team hardware
Demo sensors / microcontroller / gateway	INR 5,000–INR 20,000	Only if a physical demo is required
Prototype software	INR 0–INR 3,000	Open-source tools; optional paid API usage
Hosting / storage	INR 0–INR 2,000 per month	Local deployment or low-usage cloud tier
Indicative initial total	**INR 5,000–INR 23,000**	Excludes labour and existing equipment

These are rough budgeting assumptions to be checked against current supplier prices and the team's actual implementation. A simulation-only prototype may cost less; industrial deployment requires a separate site-specific estimate.

Use Cases

Use Case 1 – Motor bearing fault investigation

User: Maintenance technician.
Scenario: Vibration rises and a motor-temperature warning appears.
System response: Correlate available readings and alarms, retrieve bearing-related manual guidance and relevant maintenance history, then rank plausible causes and suggest safe verification checks.
Expected outcome: Faster, better-informed investigation; the technician confirms the actual fault.

Use Case 2 – Conveyor system alarm cascade

User: Automation / troubleshooting engineer.
Scenario: A conveyor stops and multiple drive, overload, or sensor alarms are reported.
System response: Group related alarms, inspect available current/speed/status readings, and surface relevant troubleshooting steps from the equipment documentation.
Expected outcome: Help distinguish a likely initiating fault from secondary alarms and reduce unstructured searching.

Use Cases

Use Case 3 – Repeated machine failure

User: Shift supervisor and maintenance engineer.
Scenario: The same machine reports a similar fault after a previous repair.
System response: Retrieve earlier work orders, replaced components, fault notes, and current symptoms; show similarities and differences for technician review.
Expected outcome: Help investigate recurring causes and avoid repeating an ineffective troubleshooting path.

Optional extension – Remote first-line troubleshooting

User: Maintenance lead supporting a difficult-to-access machine.
Scenario: A warning appears in a hot, noisy, or restricted area.
System response: Present remote sensor context, relevant documentation, and preliminary checks before an authorised person decides whether physical inspection is required.
Expected outcome: Support preparation and reduce avoidable inspections; all safety procedures remain mandatory.

These are proposed use-case scenarios, not experiments or case studies claimed by the cited research paper.

Impacts and Benefits

- **Faster investigation:** bring alarm context, relevant manuals, and historical records into one diagnostic workflow.
- **More consistent troubleshooting:** show the evidence and source references behind suggested causes and next checks.
- **Reduced disruption potential:** earlier, more focused investigation may help shorten downtime; this must be measured during testing.
- **Improved worker support:** provide remote access to relevant context before an authorised technician enters a difficult work area.
- **Knowledge retention:** make prior maintenance findings easier to find and reuse across shifts.

How we will evaluate it: time to identify a likely cause, top-k diagnostic usefulness reviewed by an expert, relevance of retrieved documents, and technician feedback. No performance improvement is claimed until tested.

Sustainable Development Goals (SDGs)

SDG	Target	Goal name	Connection to solution
SDG 9	Target 9.4	Industry, Innovation and Infrastructure	Supports reliable operations and technology-enabled industrial upgrading.
SDG 12	Target 12.5	Responsible Consumption and Production	Better diagnosis may reduce unnecessary replacement and material waste; not yet measured.

SDG alignment describes the solution's intended contribution. Actual impact depends on deployment and measured outcomes.

Individual Member Contribution

Member	Contribution / responsibility
Tharunesh P	Problem identification, industrial maintenance research, existing solution analysis, and research gap identification.
Vijeeshvar S S	Industrial IoT research, machine sensor and alarm data integration, and proposed system architecture and workflow.
Rupesh Gokula Krishnan	AI-assisted fault diagnosis research, root-cause analysis, maintenance document integration, and troubleshooting recommendation design.
Siddharth T	Solution documentation, use-case identification, expected impact and validation planning, and presentation coordination.

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